

Total No. of Questions : 8]

SEAT No. :

P-6645

[Total No. of Pages : 2

[6181]-209

B.E. (Information Technology)

AUGMENTED AND VIRTUAL REALITY

(2019 Pattern) (Semester - VIII) (Elective - VI) (414452B)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Explain different types of Visual Perception in detail. [9]

b) Explain concept of Visual Rendering in detail. [9]

OR

Q2) a) Explain concept of Motion in Real and Virtual Worlds in detail. [9]

b) Explain concept of Tracking in detail. [9]

Q3) a) How does Augmented Reality Work? Explain with suitable example. [9]

b) Describe any two applications of Augmented Reality. [8]

OR

Q4) a) Define Augmented Reality? Explain History of Augmented Reality. [8]

b) Explain Ingredients of an Augmented Reality Experience. [9]

Q5) a) Describe Sensor tracking methods in detail. [9]

b) List and Discuss different displays used in Augmented Reality systems. [9]

OR

Q6) a) Describe different types of AR devices. [9]

b) Describe sensor fusion with example. [9]

P.T.O.

- Q7)** a) What is natural-feature tracking? Describe with any one application. **[9]**
b) What is simultaneous localization and mapping in AR? **[8]**

OR

- Q8)** a) What are the major software components in augmented reality? **[9]**
b) What is marker tracking in AR? Explain in detail. **[8]**

